Fiscal Year:	FY 2021	**Task Last Updated:**	FY 01/12/2021
PI Name:	Hammond, Timothy M.D.		
Project Title:	Fuel to Mars		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	Alternate email: Timothy.Hammond@va.gov NOTE: PI previously at Tulane University until 2007 (Ed., 09/2010) Joint appointments Professor, Department of Internal Medicine, Nephrology Division, Duke University School of Medicine, and Staff Physician, Durham VA Health Care System.		
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-Artemis1 (EM1). App A: Orion (Artemis-1) (formerly Exploration Mission-1)
Start Date:	03/18/2019	**End Date:**	03/17/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	February 2020 report: None.		
COI Name (Institution):	Allen, Patricia M.S. (Veterans Health Administration) Birdsall, Holly M.D., Ph.D. (Institute For Medical Research, Inc.) Giaever, Guri Ph.D. (University of British Columbia, Canada) Nislow, Corey Ph.D. (University of British Columbia, Canada)		
Grant/Contract No.:	80NSSC19K0706		
Performance Goal No.:			

Performance Goal Text:	NOTE: End date changed to 3/17/2022 per NSSC information (Ed., 5/18/21)
Task Description:	The primary research product this project will produce is identification of the insertion strains of Chlamydomonas reinhardtii, a single-cell green alga, which optimizes survival during exposure to the combined impacts of space radiation and microgravity in the cosmic space environment beyond the Van Allen Belts. The Chlamydomonas reinhardtii strains in a mapped insertional mutant library are a transformative tool allowing near genome wide analysis of the genes mediating survival advantage during stimuli such as cosmic radiation and microgravity. The secondary set of research products this project will produce are identification of the pathways mediating the gene changes, visualization of the pathway interactions, and identification of any environments giving similar gene pathway signatures in other model systems. Our goal is to identify strains of Chlamydomonas reinhardtii with gene insertions, which convey the best survival advantage during the exposure to the combined impacts of space radiation and microgravity for future use as parent strains in hydrogen production initiatives. Hydrogen is the lead contender for fuel production for return space flight missions from orbital bodies beyond low Earth orbit. The metabolism of select microorganisms makes hydrogen fuel. The best-known and most efficient hydrogen producer is Chlamydomonas reinhardtii, which is poised for exploitation as an industrial biotechnology platform for biofuel production. The type of investigation we propose is experiments that will be conducted aboard the Exploration Mission-1 (EM-1) [Note now called Artemis] lunar orbit flight, with asynchronous controls in Kennedy Space Center ISSES (International Space Station Environmental Simulators) Chambers. The proposed hardware is the NASA Biological Research in Canisters (BRIC)-100VC. Hypothesis: The hypothesis to be tested is that the mapped insertional mutant library of Chlamydomonas reinhardtii will allow identification of the genes that mediate optimal radiation resistance, and imperviousness to microgravity, in the cosmic space environment beyond the Van Allen Belts. The specific aims of the proposal are: Specific Aim 1: To experimentally determine which of the thousands of mapped insertion mutant strains of Chlamydomonas reinhardtii have the best long-term survival advantage during combined space radiation and microgravity impacts in the cosmic space environment beyond the Van Allen Belts. Specific Aim 2: To analyze the dataset of insertion strains with the best long-term survival advantage during combined space radiation and microgravity impacts to: • Perform pathway analysis using gene set enrichment analysis (GSEA); • Visualize the pathway interactions using Cytoscape/ClueGo; • Make comparison to homologous gene sets in Chlamydomonas, Saccharomyces cerevisiae, and other model organism databases for any environments giving similar gene pathway signatures. Although focused on gene pathways for survival for future applications in hydrogen fuel production, the data generated will provide genome level insights into the cancer risks of prolonged exposure to cosmic radiation and microgravity. The spectrum of cancer risk genes is known, and this study will parse the importance of risk genes, and the patterns of radiation risk change with exposure to the cosmic space environment beyond the Van Allen Belts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research impacts and Earth benefits include: o The first opportunity of apply modern “omics” techniques to samples flown beyond the van Allen belts, and recovered for ground-based analysis. o The first biological validation of the utility of the Chlamydomonas deletion series for academic and commercial applications. o Delineation of the production of the potential fuels, lipid and hydrogen, to optimize use in future space based and ground applications. o Education of the next generation of space flight investigators. o This proposal directly addresses the Space Biology Plan Element GL3: What are the key cellular and molecular systems necessary for biological organisms to thrive in space environment that can be discovered from 21st century bioinformatics tools? (also Decadal survey page 71). o This application directly addresses three elements of the Human Research Roadmap on Risk of Radiation Carcinogenesis: Cancer 07: How can systems biology approaches be used to integrate research on the molecular, cellular, and tissue mechanisms of radiation damage to improve the prediction of the risk of cancer and to evaluate the effectiveness of countermeasures? How can epidemiology data and scaling factors support this approach? Cancer 08: What are the most effective medical or dietary countermeasures to mitigate cancer risks from exposure to SPE (solar particle events) and GCR (galactic cosmic rays)? What side effects should be tolerated versus mission risks? Can medical countermeasures that provide mitigation across multiple cancer types or across multiple radiation risk areas be identified? Cancer 09: Are there significant effects from other spaceflight factors that modify the carcinogenic risk from space radiation?

Task Progress:	Progress on the progress is characterized by completion of the Experiment Verification Test (EVT) and publication of an article documenting the properties of the hardware. EVT parameters requested by program. “For the EVT parameters, Tuesday October 20, 2020 – Monday December 14, 2020 (55 days) were allotted to perform the EVT. The objective of the Experiment Verification Test was to verify that the Experiment and hardware together will enable success based on the Experiment criteria set forth in the Experiment Requirements Document (ERD). Thus, to a large extent the work performed with Moonshot and the maturity of the hardware and Experiment will better define what is needed to be done prior to and during the Artemis 1 mission time period.” EVT Protocols. The EVT was performed at the Space Life Sciences Laboratories at Kennedy Space Center. A loop inoculum of approximately 2,000 mapped insertional mutant of Chlamydomonas reinhardtii were placed into liquid Tris- acetate-phosphate (TAP) media (1M Tris base 20 ml, phosphate K2HPO4/KH2PO4 buffer 1ml, Hutner’s Trace metals 1 ml, 10 ml of solution A [NH4Cl, MgSP4.7H2O, CaCl2.2H2O], glacial acetic acid 1 ml pH 7 made up to 1 liter with distilled water), and exposed to white light. After a lag of ~21 days, the Chlamydomonas grew robustly. TAP media plus 12 grams agar per liter was autoclaved and 10 ml poured into 100 mm diameter Petri dishes and single well Omniplates. The dark green, robustly growing, Chlamydomonas cultures were robotically spotted onto the plates. The plates were parafilmed and allowed to sit for two days, at room temperature, in a box containing red and blue light coming on simultaneously for 6 hours out of a 24- hour cycle. The plates were placed in the flight environmental simulator and the EVT initiated Tuesday October 20, 2020. The EVT was completed on Monday December 14, 2020, after 55 days. The Chlamydomonas reinhardtii mapped insertional mutant library of 1935 strains continue to be maintained on TAP agar filled agar plates, and replated robotically every two weeks. For the EVT we will robotically spot the mapped insertional mutant library. Results of the Experiment Verification The Chlamydomonas mapped insertional mutant library is maintained on TAP agar plates and aliquots reseeded onto fresh plates robotically every 2 weeks. There have been no growth or contamination problems to date over a period beyond 12 months. All Chlamydomonas success criteria were met during the EVT specifically: Grow wildtype Chlamydomonas, Grow and serially maintain all Chlamydomonas deletion clones, (Lack of) Contamination Hardware performance. The hardware performed nominally throughout the Experiment Verification Test. The power profiles validate that the lights came on for 6 out of 24 hours daily as planned. There was >25% of the power remaining in the batteries at the end of the protocol. The maximum temperature increase was 5oF, well within the envelope of tolerable experimental conditions. All hardware success criteria were met during the EVT: Temperature recording, Voltage recording, Acceleration recording, Biological sample analysis. At the end of the EVT, samples were harvested for next generation sequencing, spectrophotometric analysis of chlorophylls, and flow cytometric analysis of lipid content and the fraction of live, apoptotic, and necrotic Chlamydomonas populations. Spectrophotometry analysis. The spectrophotometric absorption of Chlamydomonas was assayed serially in cultures spotted on an Omniplate, as described above. A Styrofoam box adjacent to the spectrophotometer was rigged with cycling lights to mimic as closely as possible the EVT flight hardware. The Omniplate was moved into the spectrophotometer at intervals throughout the 445-day period and absorptions were measured at 436 nm, 472 nm, 700 nm, 554 nm, and 800 nm. All spectrophotometric success criteria were met during the EVT: Protein content, Chlorophyll, PSI and PSII, Cytochrome f. Flow cytometry. In flow cytometry analyses, we counted 2,000 Chlamydomonas, and recorded forward and side scatter in addition to fluorescent parameters on a Beckton-Dickenson Accuri C+ flow cytometer using log amplified photomultipliers. Annexin and propidium iodine flow cytometry signals were gated on an unstained sample to identify and exclude background fluorescence. Annexin reports apoptosis and propidium iodide reports cell death. All flow cytometry success criteria were met during the EVT: Assay viability, Programmed cell death, Lipid content, Cell size distribution. Sequencing analysis of Chlamydomonas samples by next generation sequencing. Genomic DNA from Chlamydomonas was extracted from 107 cells using the Qiagen DNeasy Plant Mini Kit. We added cetyl trimethylammonium bromide (CTAB) to the lysis buffer to precipitate carbohydrate which would otherwise contaminate the DNA. Approximately 5 ug of DNA was obtained and judged to be of sufficient quality by agarose gel electrophoresis. Because Next Generation Sequencing is planned for sample analysis after the actual flight, and we had access to low-cost Illumina short-read sequencing, we proceed to analyze three independent clones via Next Generation Sequencing technology in the EVT. Each of the three samples gave >130 million reads with a 95% mapping rate. This translates into 92-fold coverage of the genome which will provide for robust variant calling because each base of the 120 mB Chlamydomonas genome is sequenced, on average, 92 times. All next generation sequencing success criteria were met during the EVT: Next generation sequencing, Quality of sequencing, Sequencing analysis.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Hammond TG, Allen PL, Wells HW, Russick JM, Nislow C, Giaever G, Birdsall HH. "Moonshot: Affordable, simple, flight hardware for the Artemis-1 mission and beyond." Front Space Technol. Published: 11 November 2020. 1:593523. https://doi.org/10.3389/frspt.2020.593523 , Nov-2020

Articles in Peer-reviewed Journals	Birdsall HH, Hammond TG. "Role of shear stress on renal proximal tubular cells for nephrotoxicity assays." J Toxicol. 2021 Apr 21;2021:6643324. Review. https://doi.org/10.1155/2021/6643324 ; PMID: 33976696; PMCID: PMC8084667 , Apr-2021